

ABYLAY AMANBAYEV

📍 Merced, CA 📞 +1 (209) 500-8452 🔗 LinkedIn 🌐 GitHub ✉ UC Merced

EDUCATION

University of California Merced

Aug 2023 - Present (Exp. 2027-2028)

PhD in Computer Science — Qualifying Exam Passed (2026)

Advisor: Prof. Florin Rusu — *Focus:* Query Optimization, Vector Databases, Agentic Retrieval

HSE University, *Master of Data Science*

2021 - 2023

EXPERIENCE

University of California Merced — *Teaching/Research Assistant*

Aug 2023 - Present

- **Current work:** Designing a query optimizer for Filtered Approximate Nearest Neighbor Search and a retrieval optimizer for *agentic retrieval* — selecting which backend to query (vector, relational, lexical, etc.).
- **Filtered Vector Search:** Engineered and benchmarked Vector Databases (PG-vector, Milvus, and FAISS) across vector search indexes (including HNSW, IVF); developed the new metric to predict query latency.
- **Data Pipeline Engineering:** Architected and released *MoReVec*, a large-scale multi-modal benchmarking dataset; implemented automated data mining and ETL pipelines for embedding ingestion.
- **Coursework & Quals:** Database Systems Implementation, Advanced Algorithms, Deep Learning, Compilers; passed Qualifying Exam (2026) on Filtered Approximate Nearest Neighbor Search.
- **Teaching Assistant:** CSE 111 Database Systems, CSE 140 Computer Architecture, CSE 031 Computer Organization — led labs, designed SQL/optimizer assignments, mentored 60+ students/term.

dalel.ai — *Founder & Sole Engineer*

2026

- Built a chatbot over the Kazakhstan tax legislation: indexed the full large text corpus for exact document lookup plus Chroma vector search for semantic retrieval, returning cited answers. Demo: [letmesee / dalelai](#)
- Designed the *kaztax* MCP server (FastMCP): a model-agnostic tool layer the agentic LLM (Gemini / Claude / GPT) drives to invoke those fetch and search methods, returning cited, grounded answers.
- Engineered the full stack solo — PostgreSQL with async SQLAlchemy + Alembic, FastAPI with structlog and rate limiting, Next.js / React frontend, deployed via Railway.

Lawrence Livermore National Laboratory — *Team Lead, Data Science Challenge*

Summer 2024

- Led a 4-person team applying deep learning to biomedical signal reconstruction.

PUBLICATIONS & PREPRINTS

1. **A. Amanbayev**, B. Tsan, T. Dang, F. Rusu. "*Filtered Approximate Nearest Neighbor Search in Vector Databases: System Design and Performance Analysis.*" VLDB 2026 VecDB Workshop. arXiv:2602.11443 [cs.DB], Feb 2026.
2. B. Tsan, **A. Amanbayev**, A. Datta, F. Rusu. "*Sketched Sum-Product Networks for Joins.*" arXiv:2506.14034 [cs.DB], June 2025.
3. A. Datta, B. Tsan, Y. Izenov, **A. Amanbayev**, F. Rusu. "*PLANSIEVE: Real-time Suboptimal Plan Identification through Incremental Learning.*" arXiv:2501.16544 [cs.DB], Jan 2025.
4. Y. Izenov, A. Datta, B. Tsan, **A. Amanbayev**, F. Rusu. "*Spanning Tree-based Query Plan Enumeration.*" arXiv:2403.04026 [cs.DB], Mar 2024.

TECHNICAL SKILLS

Languages	Python, SQL, C/C++, TypeScript / JavaScript, $\text{\LaTeX}$
Databases	PostgreSQL, MySQL, SQLite, ClickHouse, Milvus, FAISS, PG-vector, ChromaDB
Infrastructure	ETL/ELT, Alembic, Docker, AWS S3, Railway, GitHub, Linux/Bash
ML & LLMs	PyTorch, TensorFlow, HuggingFace, LangChain, MCP / FastMCP, Agentic RAG
Backend	FastAPI, async SQLAlchemy, structlog, Next.js / React